Amir Ali Farzin

PhD Student, Australian National University

amirali.farzin@anu.edu.au | github.com/amirali78frz

Google Scholar
LinkedIn
+61 468583866

Professional Summary

PhD candidate at the Australian National University working on optimisation methods for modern machine learning systems. My research develops zeroth-order and saddle-point algorithms for large-scale and nonconvex min–max problems, combining theoretical guarantees with efficient implementation. I am motivated to translate rigorous optimisation insights into scalable solutions for real-world learning and decision-making problems.

Education

Ph.D. in Engineering and Computer Science
Australian National University (ANU)
June 2023 – Present
Supervisor: Prof. Iman Shames and Dr. Philipp Braun

B.S. in Mechanical Engineering
Sharif University of Technology (SUT), 2017-2021
GPA : 17.41/20 (3.78/4)

High-School Diploma
Alborz High-School, 2013-2017
GPA : 19.86/20 (4/4)

Selected Publications

A. A. Farzin, Y.-M. Pun, P. Braun, A. Lesage-Landry, Y. Diouane, and I. Shames, "Min-max optimisation for nonconvex-nonconcave functions using a random zeroth-order extragradient algorithm," Transactions on Machine Learning Research, 2025.

Farzin, A.A., Pun, Y.M., Braun, P. and Shames, I., 2025. Properties of Fixed Points of Generalised Extra Gradient Methods Applied to Min-Max Problems. IEEE Control Systems Letters (L-CSS).

Farzin, A. A., & Shames, I. Minimisation of Polyak-Łojasewicz functions using random zeroth-order oracles. In 2024 European Control Conference (pp. 3207-3212). IEEE.

Farzin, A. A., Pun, Y. M., Braun, P., Summers, T., & Shames, I. (2026). Solving the Offline and Online Min-Max Problem of Non-smooth Submodular-Concave Functions: A Zeroth-Order Approach. arXiv preprint arXiv:2601.21243.

Professional Experience

Student Researcher, ANU
June 2023 – Present
Supervisor: Prof. Iman Shames

- Solving and Formulating Partial Information Decision Problems

Peer Reviewer, International Conference on Machine Learning (ICML), IEEE Transactions on Control of Network Systems (TCNS), IEEE Transactions on Automatic Control (TAC), IEEE Control Systems Letters (L-CSS), Conference on Decision and Control (CDC), European Control Conference (ECC)

Projects

Zeroth-Order Minimisation

- Minimisation of quasar-convex functions using random zeroth-order oracles
- Minimisation of Polyak-Łojasiewicz functions using random zeroth-order oracles
- Minimisation of submodular functions using random zeroth-order oracles

Zeroth-order Min-max Optimisation

- Min-Max optimisation for nonconvex-nonconcave and possibly non-differentiable objective functions using a random zeroth-order extra gradient algorithm
- Solving the Min-Max Problem of non-smooth Submodular-Concave Functions: A Zeroth-Order Approach

Saddle Point Problems

- Properties of fixed points of generalised Extra Gradient methods applied to min-max problems
- Building a framework with guarantees to analyse the discrete-time algorithms from the viewpoint of continuous-time dynamical systems
- Analysing the state-of-the-art minimax algorithms from the perspective of convergence to saddle points
- Designing an innovative continuous flow with guarantees of convergence only to the saddle points

Raceline Optimisation

- Developing a Python package for raceline optimisation, incorporating new and state-of-the-art methods (ongoing).
- Building a dataset for raceline optimisation and training a neural network taking track map information as input and efficiently output optimised racelines and velocity profiles with high accuracy

Talks and Presentations

- University of Melbourne, 2025 – Invited talk on min-max optimisation for nonconvex-nonconcave functions using a random zeroth-order extragradient algorithm.
- IEEE European Control Conference (ECC), Stockholm, Sweden, 2024 – Contributed talk on minimisation of polyak-Łojasewicz functions using random zeroth-order oracles.
- University of Sydney, 2025 – Invited talk on min-max optimisation for nonconvex-nonconcave functions using a random zeroth-order extragradient algorithm.
- IEEE Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, 2025 – Scheduled talk on properties of fixed points of generalised extragradient methods applied to min-max problems.
- Optimisation Days at UNSW, 2025 Talk on properties of fixed points of generalised extragradient methods applied to min-max problems and beyond.

Research Interests

- Optimisation algorithms for Machine Learning (Zeroth-Order and Stochastic methods)
- Min-max optimisation for adversarial training and robust ML
- Learning-Based Control and Multi-Agent Systems
- Saddle point problems

Technical Skills

Languages: Python, MATLAB, C++
Tools: Simulink, SOLIDWORKS, Arduino, Proteus, Git
Machine Learning: PyTorch, TensorFlow, OpenCV, JAX

Teaching Experience

Teaching Assistant (TA), ANU and SUT
Sep 2022 – Dec 2025

- Control Systems (Convener: Dr. Philipp Braun), ANU, Semester 2, 2024 and 2025
- System Dynamics (Convener: Dr. Igor Vladimirov), ANU, Semester 1, 2024
- Automatic Control (Convener: Prof. Aria Alasty), SUT, Semester 1, 2022

Private Teacher
Aug 2017 - Aug 2019

- Taught physics, mathematics, and chemistry to high-school final-year students

High-School Academic Counselor, Alborz High-School
Aug 2017 - Aug 2021

- Advised over 400 students on academic planning and managing their educational pathways

Honors & Awards

- Ranked 162nd out of 150000+ participants (Top 0.1%) in Iran's National University Entrance Exam (2017)
- Selected to participate in the competitive Machine Learning Summer School (MLSS) 2026 in Melbourne.
- Exempted from Iran's university entrance exam for master's degree at Sharif University of Technology (Merit-based exemption)
- Received a full scholarship package for international Ph.D. candidates from the ANU School of Engineering.
- Ranked 5th out of 50+ teams in F1TENTH Sim Racing League at CDC 2024, using control and trajectory planning algorithms.
- Won the Student Travel Award for both the Workshop and main conference to attend the Conference on Decision and Control (CDC) 2025.
- Awarded an Australian Mathematical Sciences Institute (AMSI) Travel Grant to attend AMSI Summer School 2025.
- Top 15% of SUT entrants (2017)

Languages

English: TOEFL 106/120
Persian: Native